

Report 7 - Generator Model Family for OpenHPL/OpenHPLjl

Madhusudhan Pandey

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Abstract

This report introduces the generator model family. The first implementation is a reduced algebraic mechanical-to-electrical power converter; later models will add swing dynamics, rotor angle, voltage, and synchronous-machine states.

1 Updated Context

The generator converts shaft mechanical power into electrical active power:

$$\boxed{\text{Shaft} \rightarrow \text{Generator} \rightarrow \text{Grid}}.$$

2 Generalized Concept Model

Inputs are shaft speed and torque. The principal electrical output is active power:

$$\omega, \tau \rightarrow \text{Generator} \rightarrow P_e.$$

3 Other Models Available in the Literature

A useful hierarchy is

$$\boxed{\text{ideal converter} \rightarrow \text{swing model} \rightarrow \text{classical synchronous machine} \rightarrow \text{higher-order dq machine}}.$$

4 Mathematics Involved

Mechanical input power is

$$P_m = \tau\omega.$$

The first reduced generator uses

$$\boxed{P_e = \eta_g P_m}.$$

Later synchronous-machine models will introduce

$$M\dot{\omega} = P_m - P_e - D\Delta\omega$$

and rotor-angle dynamics.

5 OpenHPL-Specific Modeling Interpretation

OpenHPL supports turbine-generator mechanical coupling and power-system integration. OpenHPLjl begins with a transparent reduced converter before adding detailed machine equations.

6 Implementation in Julia ModelingToolkit / SciML

```
@component function IdealGenerator(; name, eta=0.98)
    @named shaft = RotationalPort()
    @named grid = ElectricalPowerPort()

    Pm ~ shaft.tau * shaft.omega
    Pe ~ eta * Pm
    grid.P ~ -Pe
    grid.omega ~ shaft.omega
end
```

7 Model Assembly and Connections

The shaft rotational port connects to the generator mechanical port; the electrical power port connects to an infinite-grid or future dynamic-grid model.

8 Numerical Experiment / Validation

For

$$\tau = 1000 \text{ N m}, \quad \omega = 100 \text{ rad s}^{-1}, \quad \eta_g = 0.98,$$

$$P_m = 100 \text{ kW}, \quad \boxed{P_e = 98 \text{ kW}}.$$

9 Expected Outputs

The minimum outputs are

$$P_m(t), \quad P_e(t), \quad \omega(t).$$

10 Engineering Interpretation

This reduced model is useful for early energy-balance and frequency-control assembly. Detailed synchronous-machine states should only be added when voltage and rotor-electromagnetic dynamics are required.

Conclusion

The generator family now has a clean starting interface. Next models are a swing-equation generator and a classical synchronous generator.